BitSaver: Recovering BitLocker Partitions

Alvin Leung
Computer Science and Engineering
Univerisity of Nevada, Reno
Reno, United States of America
aleung@unr.edu

***Abstract*—Due to the rise of cybercrime and data theft, industries and consumers have started worrying about their data and computing practices. This has led to a majority of industries and homes to begin practicing data encryption. Window's native BitLocker system provides enterprise and home users with an easy option to safely encrypt and decrypt their files, discouraging data theft. Thus, BitLocker disks are becoming more and more common in both law enforcement and digital forensic fields. However, likewise, accidental partition deletion with BitLocker is also just as common. Whether accidental or intentional, data recovery has become complicated, especially with BitLocker. With a BitLocker disk, standard troubleshooting and data recovery techniques are not sufficient due to BitLocker encrypting the data into an unreadable format that no human nor software can read without a decryption key. However, BitSaver is a new command-line interface (CLI) program that detects and finds BitLocker partitions and rebuilds partition tables to allow an encrypted disk to be readable again, assuming that a BitLocker recovery key exists and is in the possession of law enforcement or data forensics .**

***Index Terms*—encryption, data, forensics, BitLocker, recovery.**

I. Introduction

A. BitLocker and Digital Forensics

BitLocker is an application that is very relevant in the field of Information Technology (IT) and digital forensics. Most enterprises have policies that allow their systems to automatically and routinely encrypt data. On top of this, the enterprise may even recycle keys monthly to further protect data. This is to prevent confidential data from being stolen. However, this often creates new barriers for digital forensics teams when they need to analyze a disk for official reason such as law enforcement. According to North Forensics, encryption makes data unreadable without a key, bypassing even the most advanced forensic tools. Thus, BitLocker becomes a time-consuming, unavoidable, and complex process for data forensic technicians to overcome, oftentimes impossible without sophisticated tools and procedures.

On top of this, the legal aspects of evidence that utilizes encryption can be complicated. Meticulous documentation is required to maintain integrity of an encrypted piece of evidence so that it can be used in the court of law [5].

B. BitSaver Objectives and Motivation

When an individual manually deletes a BitLocker partition, the data does not get deleted. Instead, the disk marks the partition as unallocated space so that new data can be written to the unallocated partition. Thus, the data remains on the drive so long as no additional data overwrites the unallocated data. However, the data partition still cannot be read directly, either with or without data recovery software, due to the encryption from BitLocker. Suspects who are informed on disk partitioning schemes and BitLocker may delete a BitLocker partition in order to stall law enforcement or obscure any evidence that a disk might contain. This leads to law enforcement being unable to use standard data recovery procedures.

The goal of this project, BitSaver, is to create a tool that can automate the recovery of BitLocker partitions that have been deallocated into unallocated space on a disk. Usually, deallocated space naturally occurs when a storage device has not been used or formatted prior. However, it can also occur when an individual manually deletes disk partitions such as with the Windows built-in DiskPart tool. BitSaver is focused on creating a tool that works for Windows as the Windows operating system is the primary operating system that utilizes BitLocker. Thus, BitSaver does not work for other operating systems like MacOS or Linux/UNIX and will not work even in future versions.

In general, law enforcement, digital forensics, and enterprise IT technicians will be able to use this application to recover BitLocker partitions that have been deleted and not formatted (and not overwritten). For example, if a suspect knows that law enforcement is on their "trail", the suspect may delete BitLocker partitions and not format the deleted partitions of their devices to obscure the data but still be able to potentially recover the data at a later point (post-investigation). The suspect would still need to hide their recovery keys, but a barrier to the potentially vital information has formed for law enforcement. Another example would be if a trainee IT technician accidentally deleted a BitLocker partition on an important, critical data drive while trying to install the Windows operating system for another disk or partition. In these cases, law enforcement and the IT technician would have a way of potentially restoring the otherwise unrecoverable data. Of course, there are already many techniques for rewriting boot records to restore functionality. However, most tools require the user to perform potentially destructive disk writes, which can potentially cause permanent data loss. This is where the application can be very useful. BitSaver automatically detects and locates BitLocker partitions, accurately calculates partition sizes, and safely writes boot records for its users to avoid dangerous writes.

C. BitSaver Plans and Implementation

To implement this tool, standard Python libraries, PowerShell commands, and Runtime Software's open source, free DriveDoppel executable were used [7]. Python was chosen to implement a user-friendly interface to communicate with the user and to wrap special system calls. Standard libraries like Hashlib, Ctypes, etc. were used for their ability to analyze disk data and make calls to the Windows operating system. Non-standard libraries include: pyuac, pypiwin32, PySide6, and hexdump. PowerShell was used to communicate system calls to the disk safely in Windows. Lastly, DriveDoppel was used to safely write boot records to a disk.

When ran, the program will open up an interface that allows users to select numbers representing the user's choices. Initially, the program will collect information about all storage devices detected on the system, including internal and external storage devices. Afterwards, the device will allow users to perform operations on disks such as selecting disk, hexdumping BitLocker sectors, reading the boot sector, and repairing BitLocker master boot records. The system is also intentionally configured to stop users from selecting the boot drive as an option to prevent critical damage on the operating system running the program. A depiction of the menu is shown in Fig. 1. Users in this screen are able to see the hostname of the machine they are working on and see many useful, surface-level attributes about the storage devices connected to the machine such as drive number, friendly drive names (official models, which is useful for documentation purposes), encryption status, and capacity.

```
Computer Name: DESKTOP-L2Q5G34

disk_number: 0
size_bytes: 1000204886016
bus_type: NVMe
model_name: Samsung SSD 970 EVO Plus 1TB
is_system: True
is_boot: True
is_readonly: False
is_removable: False
partition_style: GPT
letters: CE
volume_type: Operating System
volume_status: Fully Decrypted
capacity_GB: 930.677734


disk_number: 1
size_bytes: 15728640512
bus_type: USB
model_name: ASolid USB
is_system: False
is_boot: False
is_readonly: False
is_removable: False
partition_style: MBR
letters:
volume_type: Unknown
volume_status: Unknown
capacity_GB: 0.0



===Main Menu===
1. List Disks
2. Select Disk
3. Exit
Enter choice:
```

Fig. 1. BitSaver CLI for users to choose options.

II. Related Work

A. Commerical BitLocker Restoration Software

Restoring deleted BitLocker partitions is not a rare troubleshooting issue. In fact, there are many online sources that provide methods to resolve such issues such as those from M3 Software or EaseUS [3], [4]. These sources all describe the relatable issue of accidentally deleting BitLocker partitions. However, almost every single one of these sources do not provide systematic command calls or low-level steps to resolve the issue.

Instead, almost every source provides their own sponsored, paid software to resolve this issue. On the other hand, BitSaver believes that potential critical data loss should not be left to the mercy of paid tools. Instead, BitSaver seeks to provide the foundation of open source software that can be used to protect and restore critical data.

B. Manual Processes

There are several sources that provide manual steps to resolve this issue, which this entire project is based on. For example, Norman Bauer, published a blog page in 2013 that instructs people on how to restore a deleted BitLocker partition using only the Windows OS and the open source 7-zip application. However, his methods merely takes a copy of the data off of a drive rather than restore the drive's original BitLocker functionality, which may have different legal implications due to evidence validity. BitSaver intends to restore a drive's BitLocker functionality without destroying the data [2].

Another source, which provides the groundwork for this application, is Runtime Software's BitLocker data recovery guide. The guide instructs the user on how to explore a disk's sector to map out the partition needed to be written to the disk's partition table using DiskExplorer X, DriveDoppel, and GetDataBack Pro. While the content is very well-written, the guide still attempts to sell its paid software GetDataBack Pro. However, its freeware DiskExplorer X was used in this project to verify the integrity of each step taken by the application in

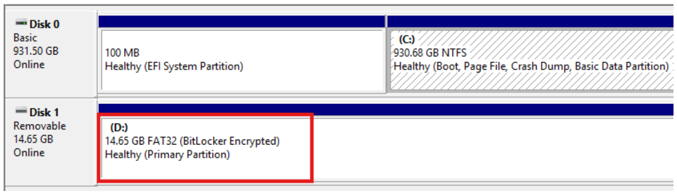

Fig. 2. BitLocker encrypts the 16 GB USB drive, and Windows recognizes the BitLocker properties of the USB drive.

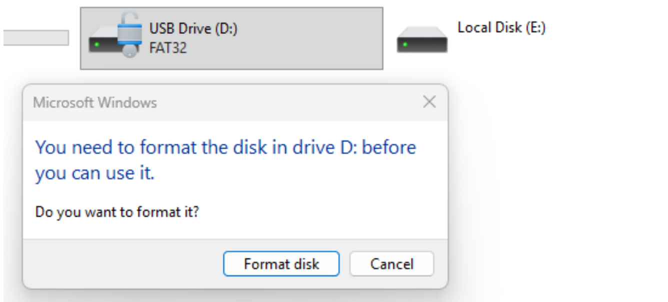

Fig. 3. DiskPart operation renders the drive inaccessible, which prompts user to reformat the disk.

order to make sure that the application is performing correct computations and searches [6]. Furthermore, DriveDoppel is directly used in the program for the accurate MBR writes. Lastly, GetDataBack Pro was used to verify the inaccessibility of the data stored on deleted, locked BitLocker partitions [1].

III. Methodology

A. *Gathering Disk Information*

To start off, an easy and reproduceable testing environment is created. In this environment, a small 16 GB USB drive was used as the primary form of storage device in BitSaver's earliest tests. Due to the time it takes to encrypt and decrypt a large drive, larger drive sizes were not considered for initial testing of functionality. The 16 GB USB drive was formatted to FAT32 and the recovery key was stored externally for unlocking and verifaction. To begin, Windows's built-in BitLocker application was used to encrypt the 16 GB USB drive as shown in Fig. 2.

Afterwards, the disk's BitLocker partition was manually deleted via Windows DiskPart. This resulted in Windows Disk Manager recognizing the partition as unallocated as shown in Fig. 4. Accessing the data also prompts a formatting request (which must not be accepted to prevent data loss) as shown in Fig. 3. Standard data recovery software yields no results when files are analyzed for recovery such as Get Data Back Pro (GDBP).

When a partition deletion occurs, the disk's partition table has been modified to mark the deleted partition as unallocated. A rewrite of the boot record on the disk is required to restore functionality to the disk (essentially to undo the operation preceding the deletion. However, the boot record is incredibly

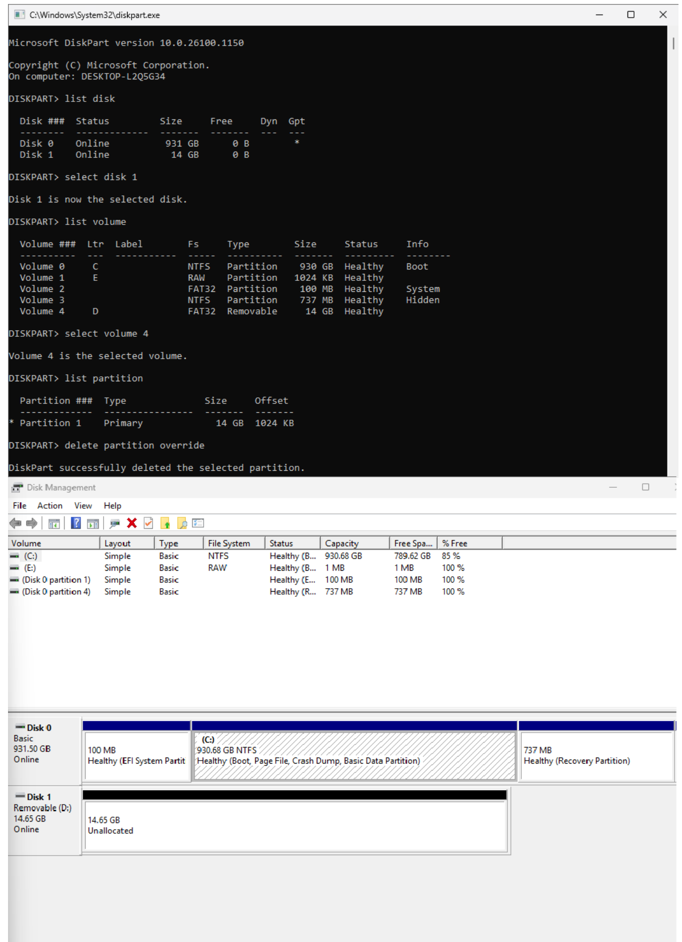

Fig. 4. DiskPart operation to forcefully delete BitLocker partition (Top). Disk becomes unallocated after deletion (Bottom).

sensitive and requires an accurate byte count of the size of the BitLocker partition to avoid deleting unintended data.

Thus, the first step required is to gather the data of all storage devices to obtain information about the partitions on each disk. The program utilizes the Windows built-in PowerShell command Get-BitLockerVolume and Get-Disk to gather and format data regarding all data storage on the system. Some of the most important information gathered about the disks is the disk number, size, boot status, drive letters, and BitLocker status. All of this information is neccessary to perform accurate, informed disk operations.

Afterwards, BitSaver marks the disk that the program is running off as unprocessable and unselectable to protect the native boot drive from destructive disk writes. If selected, the program will error out and prompt user to choose a different disk. All other devices are allowed to be operated on. If a valid, non-boot device is selected, deeper disk information is gathered via the Windows Get-Disk function such as logical sector size, Logical Block Addressing (LBA) sector count, and true size. Logical sector size is the total size of a single sector measured in bytes. The LBA sector count is the number of logical sectors on the storage device. Lastly, the total size is

Fig. 5. BitSaver calculates the logical sector size, LBA sector count, and size of the disk selected for disk operations.

the number of bytes that the device can hold. The LBA sector count can be calculated by the following:

$$s/l = \gamma \tag{1}$$

where s is the capacity of the storage device in bytes, l is the logical sector size also in bytes, and γ is the LBA sector count. The LBA sector count is crucial to performing correct disk operations as a disk is accurately divided by these sectors.

B. *Finding BitLocker Sector and Count*

Using the Equation 1, it becomes trivial to calculate the number of LBA sectors for the partition table parameters as shown in Fig. 5. The next step is to calculate the start sector and the end count. The starting sector of all BitLocker partitions always has a hex pattern of ”2D4656452D46532D”. This is because all BitLocker partitions have the signature ”-FVE-FS-” [1]. The hex pattern is merely a conversion of the string ”-FVE-FS-” in hex. To find the starting sector of a BitLocker sector, BitSaver iterates from sector 0 (start of the physical disk) and finds the first iteration of the BitLocker signature in hex. In most cases, the BitLocker signature will be on sector 2048 as shown in Fig. 6.

Fig. 6. BitSaver performs a hexdump of all sectors to look for the unique, BitLocker signature.

Afterwards, the partition size needs to be calculated by the following:

$$s - b = p \tag{2}$$

where s is the capacity of the storage device in bytes, b is the starting sector for the BitLocker partition, and p is the partition size. This is used to calculate the partition size to write to the partition table in the next step.

C. *Rebuilding the BitLocker MBR*

After choosing to rebuild the BitLocker MBR, the user is prompted for the file system type of the expected drive as shown in Fig. 7. Though this information is often not known, in most cases, NTFS and FAT32 are safe choices for the Windows operating system. After the file system is selected, the application will seek out the BitLocker starting sector and the BitLocker partition size using the aforementioned equations.

Afterwards, the application calls on DriveDoppel, a third-party executable used for the sensitive disk operations. The application sends DriveDoppel its parameters (the file system type, BitLocker starting sector, and partition size) and runs the executable on the selected disk. The CLI notifies the user of the potentially destructive operation.

If the user approves, DriveDoppel writes the new partition data to the partition table as shown in Fig. 8. With a correct write, Windows should recognize the disk as a disk with

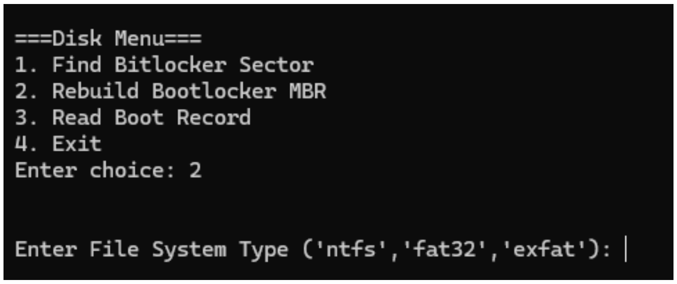

Fig. 7. BitSaver prompts the user to enter the expected file system of the disk.

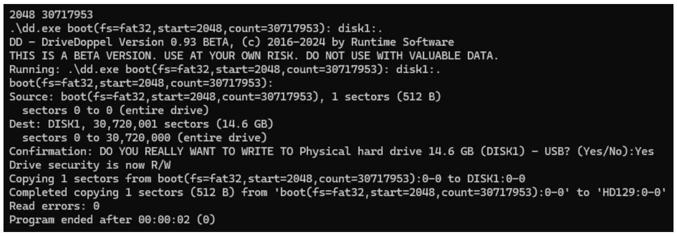

Fig. 8. DriveDoppel running on a disk and writing a FAT32 file system with parameterized sector values.

BitLocker enabled and prompt the user to enter the recovery key for the disk (in some cases, a physical disconnect and reconnect may be needed for Windows to recognize the change in the device's partition table). After entering the correct recovery key, the user should be able to access the data previously locked anyway by BitLocker and unallocated partitions as shown in Fig. 9.

IV. Experimental Setup

Due to the potential for destructive disk writes on boot drives, the experimental program was installed on a computer with a minimal Windows 11 operating system (version: 25H2). If there is interest in recreating the experiment. It is highly recommended to use a device that does not have important data located on the boot drive or to use a virtual machine instead.

For the experiment to test out the integrity of BitSaver, two 16 GB USB drives (identical to the ones used during methodology), a 500 GB solid state drive, and a 256 GB non-volatile memory express (NVME), each with unique BitLocker recovery keys and encryption, were used. For the 500 GB and 256 GB devices, a test on whether or not an operating system drive can be restored was also added.

Firstly, all devices had filler data stored on them. Afterwards, BitLocker was turned on for all four individual devices. Afterwards, all four devices had their BitLocker partitions deleted through DiskPart. BitSaver will attempt to restore BitLocker functionality to each device individually. If the device does not seem to respond to the fix, a manual removal and reconnection will be applied due to occasional read issues from the intermediate reading device or unregistered, modified partition tables.

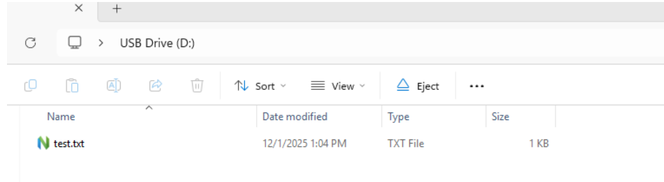

Fig. 9. Drive can be accessed again after entering recovery key.

For each device, several metrics were measured: the type of drive (data or OS), whether or not the data was recovered, capacity of the disk in bytes, number of LBA sectors, and logical sector size. These will be used to find out how effective the program is at restoring BitLocker functionality on various types of drives and storage sizes.

V. Results

A. 16 GB USB Drive, NTFS/FAT32, Data Only

The 16 GB USB had a total of 15,728,640,512 Bytes. The device only had a data partition (FAT32 and NTFS) for the entirety of the drive. The disk had a count of 3,072,001 LBA sectors and a sector size of 512. After the operation, the data was recovered after entering the BitLocker password. The results worked for both NTFS and FAT32 drives. Interestingly, rewriting the drive as either FAT32 or NTFS worked despite the drive's initial type. However, this will not work if certain files on the drive are larger than 4 GB due to FAT32's file size limit. Alternatively, NTFS supports up to 4 EB (exabytes) for each file.

B. 256 GB NVME, NTFS/FAT32, Data Only

The 256 GB NVME had a total of 250,051,221,504 Bytes. The device only had a data partition (FAT32 or NTFS) for the entirety of the drive. The disk had a count of 488,381,292 LBA sectors and a sector size of 512. After the operation, the disk originally was not reading. However, after a reconnect, the testing operating system recognized the disk's new properties. The data was then recovered after entering the BitLocker password. The results worked for both NTFS and FAT32 drives. Similar to the previous experiment, partitioning the drive as FAT32 or NTFS worked.

C. 256 GB NVME, NTFS/FAT32, OS Drive

The 256 GB NVME had a total for 250,051,221,504 Bytes. The device had an operating system partition (Windows 11) in the middle of the drive with a System EFI partition on its left and a recovery partition on its right. The disk had a count of 488,381,292 LBA sectors and a sector size of 512. After the operation, the disk was not reading. However, even after the reconnect, the data was not recovered. Instead, the operating system interpretted the the partition as RAW. Thus, BitSaver was unable to recover data on this drive.

D. 500 GB SSD, NTFS/FAT32, Data Only

The 500 GB SSD had a total of 500,102,443,008 Bytes. The device only had a data partition (FAT32 or NTFS) for the entirety of the drive. The disk had a count of 976,762,584 LBA sectors and a sector size of 512. After the operation, the disk originally was not reading. However, after a reconnect, the testing operating system recognized the disk's new properties. The data was recovered after entering the BitLocker password. The results worked for both NTFS and FAT32 drives.

E. 500 GB SSD, NTFS/FAT32, OS Drive

The 500 GB SSD had a total for 500,102,443,008 Bytes. The device had an operating system partition (Windows 11) in the middle of the drive with a System EFI partition on its left and a recovery partition on its right. The disk had a count of 976,762,584 LBA sectors and a sector size of 512. After the operation, the disk was not reading. However, even after the reconnect, the data was not recovered. Instead, the operating system interpretted the the partition as RAW. Thus, BitSaver was unable to recover data on this drive.

VI. Conclusions

A. Findings

The program worked well over all the DATA only drives, which are the most common drives that suspects can encrypt on-the-fly. In all three experiments considering data only drives, BitSaver was able to recover the data on all of them. Some, however, required a reconnect for the testing operating system to register the drive's new partition table. In addition, it was discovered that FAT32 drives could be rewritten as NTFS drives and vice versa for this experiment. This is likely due to the fact that the FAT32 file systems can be read by all operating systems. However, due to the limitations of the FAT32 system in terms of file sizes, it is not recommended to try and convert a drive to a different file system than its original system for risk of data loss. Luckily, the program can rewrite file system types. If FAT32 does not work for a system, a user may possibly run the program again and select NTFS instead, which should prevent any data loss.

However, BitSaver did not work for any of the OS drive experiments. Every experiment resulted in a RAW partition in Disk Manager. Considering that the disk no longer recognizes the last partition and the first partition (recovery partition and EFI System partition, respectively), the program likely still needs to account for partitions that surround the BitLocker partition. In the future, the program should ignore partitions that are not within the BitLocker scope in order to potentially be compatible with BitLocker OS drives.

Overall, the size of a drive is demonstrated to not deter BitSaver's ability to recover BitLocker functionality even after an overrided partition deletion through. The only barrier to the program's effectiveness is whether the drive is an OS drive or a data drive (which means if there are more partitions on the drive than just the BitLocker partition).

B. Closing Remarks

In summary, BitLocker as a technology, though intended to secure data for enterprise and privacy reasons, has become a barrier for law enforcement and people involved in digital forensics. Even worse, disk partitions with BitLocker encryption can be manually deleted, which causes standard data reading techniques to be ineffective in reading potentially important data on a suspect's devices. Thus, suspects can take advantage of these operations and obscure important pieces of evidence from law enforcement and even maliciously recover the data at a future date. BitSaver was made to solve this problem. Through only open source and free software, BitSaver is a program that automatically detects BitLocker partitions and attempt to rewrite partition tables in order to restore BitLocker functionality on an otherwise unreadable disk. Though there are many commerically avaiable products, all of the products require money. For the processes that teach users how to write their own partition tables, the processes are complicated and not simple for the average user. Making mistakes while following instructions can lead to catastrophic data loss. Through understanding the structure of file systems and disks, BitSaver identifies BitLocker sectors (through BitLocker's unique signature) and writes new partition tables to reestablish deleted partitions and restore the disk's ability to be interpretted by the testing operating system. The results indicate great success with restoring data drives and no success with os drives. However, as the program is still in development, further changes can be applied to fix this issue.

C. Future Plans

In the future, the project would like to directly explore creating its own partition writing component without the use of third-party tools like Runtime Software's DriveDoppel. There is ground work laid out already on writing disk partition tables through hexdumps and modifying a disk on the system level. However, this capability was not explorered in this paper. In addition, the project would like to explore converting the CLI-based operations into Application Programming Interface based operations or Graphical User Interface based operations. However, for the current project scope, the CLI is believed to be sufficient to demonstrate a proof of concept. Currently, BitSaver cannot restore OS drives. However, this can be remedied in the future given the critical results of the paper. Potentially, allowing the user to select partitions within a drive would allow for the restoration to work on a partition level without affecting other partitions (similar to DiskPart). Thus, deeper menus to look into partitions would be another likely implementation.

Acknowledgement

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sector searching, and DriveDoppel, which was used for the program's partition table modifications.

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